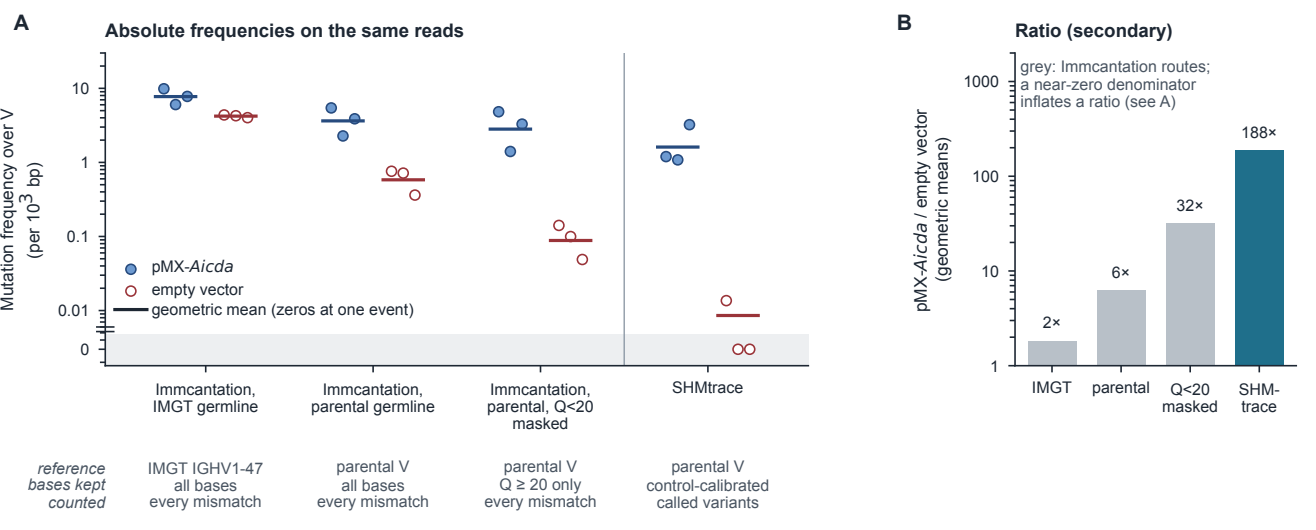

A
Absolute frequencies on the same reads
Mutation frequency over V
(per 10^3 bp)
10
1
0.1
0.01
0
pMX-Aicda
empty vector
geometric mean (zeros at one event)
Immcantation, IMGT germline
Immcantation, parental germline
Immcantation, parental, Q<20 masked
SHMtrace
reference bases kept counted
IMGT IGHV1-47 all bases every mismatch
parental V all bases every mismatch
parental V Q ≥ 20 only every mismatch
parental V control-calibrated called variants
B
Ratio (secondary)
grey: Immcantation routes; a near-zero denominator inflates a ratio (see A)
pMX-Aicda / empty vector (geometric means)
1000
100
10
1
2×
6×
32×
188×
IMGT
parental
Q<20 masked
SHM-trace